

Subject:
Physics

System:
Fluids

High altitudes are dangerous due to the low temperatures, low air pressure, and harsh weather conditions. Air pressure decreases with increasing altitude, reducing the partial pressure of oxygen in the air. Any altitude higher than 1,500 m may require medical treatment, and altitudes over 8,000 m cannot sustain human life without supplemental oxygen. Typical physiological responses to low-oxygen environments include hyperventilation and increased blood pressure. The altitude and air pressure for several locations are listed in Table 1.

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Tucson	700	700

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Table 2 Experimental Results from Hikers Pulling Sleds on Level Snow

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Some medicines used to treat altitude sickness cause dilation of blood vessels, such as the carotid artery. Which of the following changes will occur to the blood flowing through the carotid artery when it dilates?

- ☐ A. The blood velocity will increase. [21%]
- ✓ ☒ B. The blood velocity will decrease. [75%]
- ☐ C. The blood density will increase. [1%]
- ☐ D. The blood density will decrease. [1%]

Correct

76% answered correctly

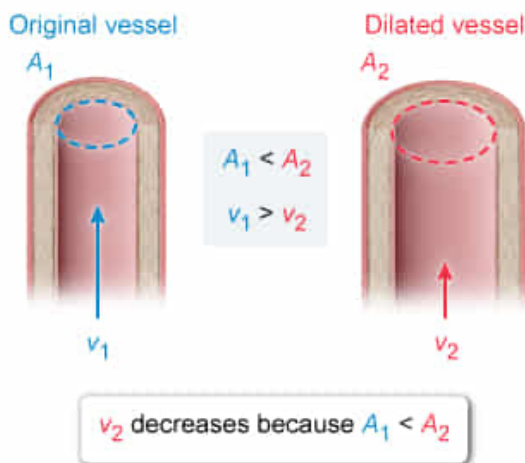
Explanation:

Blood velocity in dilated blood vessel

$$A_1 v_1 = A_2 v_2$$

$$v_2 = v_1 \frac{A_1}{A_2}$$

- A Cross-sectional area
- v Blood velocity



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The relationship between the **velocity** v of an **incompressible** fluid flowing in a pipe and the **cross-sectional area** A of the pipe is given by the **continuity equation**:

$$A_1 v_1 = A_2 v_2$$

When A changes from A_1 to A_2 , the new value of v is given by:

$$v_2 = v_1 \frac{A_1}{A_2}$$

If A_1 is less than A_2 (ie, A is increased), the above equation shows that v_2 will be less than v_1 because the ratio of A_1 and A_2 is less than one:

$$A_1 < A_2 \therefore v_1 > v_2$$

In this question, the carotid artery **dilates**, meaning A increases. Blood is assumed to be an incompressible fluid and follows the continuity equation, implying that an increase in A leads to a decrease in v . Therefore, the blood v decreases when the

carotid artery A increases.

(Choice A) Dilation of the carotid artery increases the cross-sectional area and causes the velocity to decrease, not increase.

(Choices C and D) Because blood is assumed to be an incompressible fluid, changes in the cross-sectional area of a blood vessel will not change the density of blood.

Educational objective:

The continuity equation implies that the product of blood velocity and vessel cross-sectional area remains constant. Therefore, dilation of a blood vessel will cause a decrease in the blood velocity.

Time spent:0 sec

Subject:

Physics

QID:403940

System:

Fluids

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What is the ratio of standard atmospheric pressure to the air pressure at the summit of Kilimanjaro?

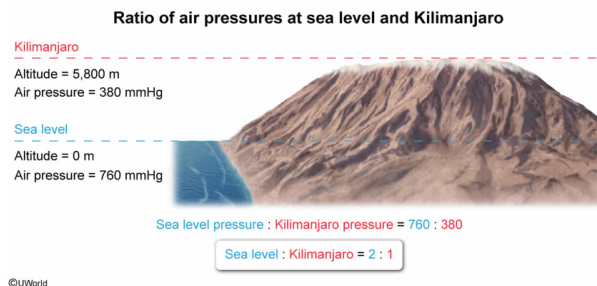
✔ ☒ A. 2:1 [88%]

- ☐ B. 3:1 [6%]
- ☐ C. 5:1 [3%]
- ☐ D. 10:1 [1%]

Correct

87% answered correctly

Explanation:



Standard atmospheric pressure (760 mmHg) refers to the air **pressure** at sea level. Air pressure is caused by the weight of the air in the atmosphere pushing down on the surface. Air pressure decreases at higher altitudes because less air is above and pushing down.

In this question, [Table 1](#) lists the air pressure at the summit of Kilimanjaro as 380 mmHg. Hence, the ratio of the air pressures at sea level and the peak of Kilimanjaro is given by:

$$\text{Sea level : Kilimanjaro} = 760 \text{ mmHg} : 380 \text{ mmHg} = 2 : 1$$

Therefore, the ratio of standard atmospheric pressure to the air pressure at the summit of Kilimanjaro is 2:1.

(Choices B, C, and D) Ratios of 3:1, 5:1, and 10:1 imply the air pressure at the summit of Kilimanjaro is 250 mmHg, 150 mmHg, or 76 mmHg, respectively, but the air pressure is given as 380 mmHg.

Educational objective:

Standard atmospheric pressure (760 mmHg) is the air pressure at sea level. At high altitudes, the air pressure decreases because less air is above and pushing down.

Time spent: 0 sec

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Physics

QID: 403941

System:
Fluids

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A hiker travels by airplane from Denver to Tucson. Which of the following will result in the same work done by gravity on the hiker?

- ☐ A. Traveling from Tucson to Denver [18%]
- ☐ B. Traveling from Pikes Peak to Tucson [1%]
- ☐ C. Traveling from Denver to Salt Lake City [4%]

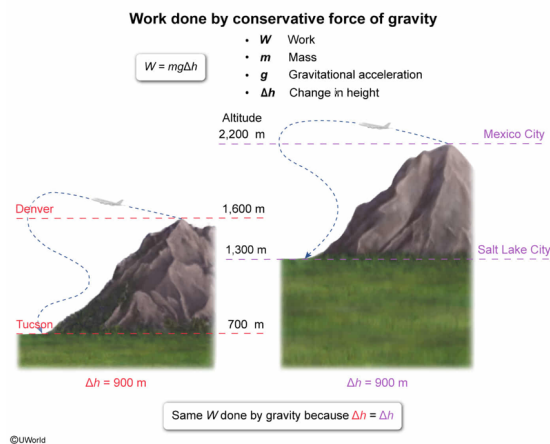


- ☒ D. Traveling from Mexico City to Salt Lake City [76%]

Correct

75% answered correctly

Explanation:



Work W equals the product of the applied force F and the displacement d . W is positive when F and d are in the same direction but negative when F and d are in opposite directions. Furthermore, when F and d are perpendicular, W is zero. For gravity, F is the product of the object's mass m and gravitational acceleration g (ie, the object's weight):

$$F = mg$$

Because g only acts downward, any horizontal components of d yield zero W done by gravity. In addition, gravity is a conservative force, so W is independent of path and depends only on the change in height Δh . Therefore, W done by gravity equals the product of m , g , and Δh :

$$W = mg \Delta h$$

In this question, the hiker travels from Denver to Tucson. Table 1 shows the hiker traveled from an altitude of 1,600 m to 700 m. Therefore, gravity does positive W because g and d are in the same direction. Furthermore, the W done by gravity is given by:

$$W = mg(1,600 \text{ m} - 700 \text{ m}) = mg(900 \text{ m})$$

Similarly, the W done by gravity when traveling from Mexico City (altitude of 2,200 m) to Salt Lake City (altitude of 1,300 m) is:

$$W = mg(2,200 \text{ m} - 1,300 \text{ m}) = mg(900 \text{ m})$$

Therefore, gravity does the same W when the hiker travels from Mexico City to Salt Lake City.

(Choice A) Traveling from Tucson to Denver results in the same magnitude of work, but the work done by gravity is negative because gravity and the displacement are in opposite directions.

(Choice B) Traveling from Pikes Peak to Tucson results in more work done by gravity because the change in height is 3,500 m.

(Choice C) Traveling from Denver to Salt Lake City results in less work done by gravity because the change in height is 300 m.

Educational objective:

Gravity is a conservative force, and the work it does on an object is independent of the object's path. Therefore, the work done by gravity depends only on the change in the object's height.

Time spent:0 sec**Subject:**
Physics**QID:**403942**System:**
Fluids

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A 10 kg sled is pulled on level snow at an average speed of 0.5 m/s. What is the average power required to pull the sled?

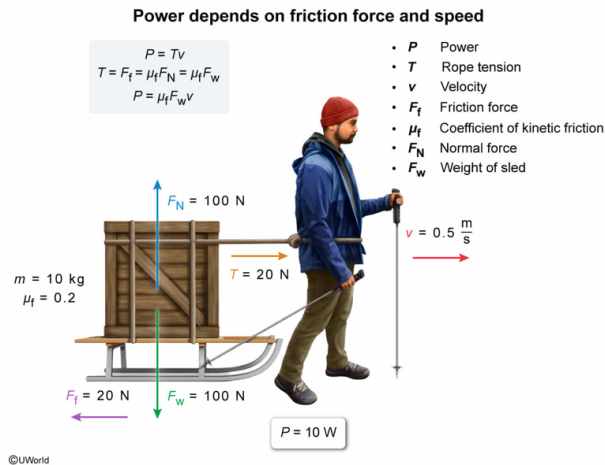
✔ ☒ A. 10 W [26%]

- ☐ B. 20 W [26%]
☐ C. 40 W [8%]
☐ D. 50 W [38%]

Correct

26% answered correctly

Explanation:



Power P is defined as the **work** W done per unit time t :

$$P = \frac{W}{t}$$

Furthermore, W equals the product of a force F acting over a distance d :

$$W = Fd$$

Combining these equations shows that P equals the product of F and velocity v :

$$P = \frac{Fd}{t} = F \frac{d}{t} = Fv$$

A sliding object experiences a **kinetic friction force** F_f equal to the product of the coefficient of kinetic friction μ_f and the **normal force** F_N that the surface exerts on the object:

$$F_f = \mu_f F_N$$

For an object on a horizontal surface, F_N equals the product of the object's mass m and gravitational acceleration g :

$$F_N = mg$$

Therefore, F_f equals:

$$F_f = \mu_f mg$$

In this question, [Table 2](#) lists values of F_f and m for different sleds. The ratio of these values in each case equals 2 N/kg:

$$\frac{F_f}{m} = \frac{30 \text{ N}}{15 \text{ kg}} = \frac{40 \text{ N}}{20 \text{ kg}} = \dots = 2 \frac{\text{N}}{\text{kg}}$$

Therefore, μ_f can be calculated as:

$$\mu_f = \left(\frac{F_f}{m}\right)\left(\frac{1}{g}\right) = \frac{2 \frac{\text{N}}{\text{kg}}}{10 \frac{\text{m}}{\text{s}^2}} = 0.2$$

Consequently, the F_f on a 10 kg sled equals:

$$F_f = (0.2)(10 \text{ kg})\left(10 \frac{\text{m}}{\text{s}^2}\right) = 20 \text{ N}$$

Because v is a constant 0.5 m/s, the pulling force on the sled (ie, the rope [tension](#) T) equals F_f , and the value of P is given by:

$$P = (20 \text{ N})\left(0.5 \frac{\text{m}}{\text{s}}\right) = 10 \text{ W}$$

Therefore, an average P of 10 W is required to pull a 10 kg sled at 0.5 m/s.

(Choice B) A power of 20 W is required to pull a 20 kg sled at 0.5 m/s, but the mass of the sled in this question is 10 kg.

(Choice C) 40 W of power is required to pull a sled with 40 N of friction force at 1 m/s, but the sled in this question has 20 N of friction force and a velocity of 0.5 m/s.

(Choice D) A power of 50 W is calculated from the product of the normal force and the velocity, but power equals the product of the kinetic friction force and the velocity.

Educational objective:

Power is equal to the product of force and velocity. Sliding an object at constant velocity requires applying a force equal but opposite to the kinetic friction force acting on the object.

Time spent: 0 sec

Subject:
Physics

QID: 403943

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Which of the following statements best explains the health risk of high systolic blood pressure at high altitudes?

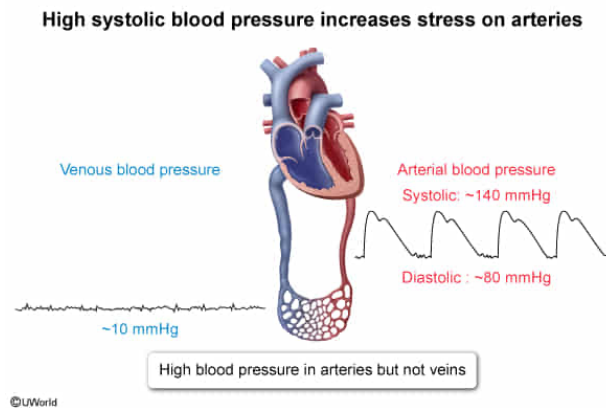
- ☐ A. High systolic blood pressure decreases the cardiac output. [8%]

- ☐ B. High systolic blood pressure decreases the blood velocity. [8%]
- ☐ C. High systolic blood pressure increases the stress on veins. [9%]
- ✓ ☒ D. High systolic blood pressure increases the stress on arteries. [74%]

Correct

74% answered correctly

Explanation:



Blood travels through vessels to deliver oxygen and nutrients to tissues and to remove waste products. Blood leaves the **heart** through the **arteries**, exchanges nutrients and wastes with tissues in the **capillaries**, and returns to the heart through the **veins**.

Arterial blood pressure increases every time the heart contracts and decreases when the heart relaxes. The maximum and minimum arterial blood pressures are called the **systolic and diastolic** pressures, respectively. However, heart contractions do not affect the blood pressure in the capillaries or veins, leading to a lower and constant blood pressure in these vessels compared to arteries.

In this question, the systolic blood pressure increases at high altitudes. Therefore, the **pressure** and stress on the arteries increases as the heart forces blood into the arteries during each contraction.

(Choices A and B) High blood pressure would increase the cardiac output and blood velocity, not decrease these quantities.

(Choice C) The systolic blood pressure is the blood pressure within the arteries, not the veins.

Educational objective:

Systolic blood pressure is the blood pressure within the arteries when the heart contracts. One health risk of high systolic blood pressure is increased stress on the arteries.

Time spent:0 sec

Subject:

Physics

QID:403944

System:

Fluids

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A hiker applies 300 N of tension on a vertical rope attached to a 50 kg box of supplies at rest on the ground while attempting to lift the supplies up a sheer cliff. What is the magnitude of the normal force the ground exerts on the box?

- ☐ A. 500 N [28%]
- ☐ B. 300 N [11%]
- ☒ C. 200 N [54%]
- ☐ D. 0 N [4%]

Correct

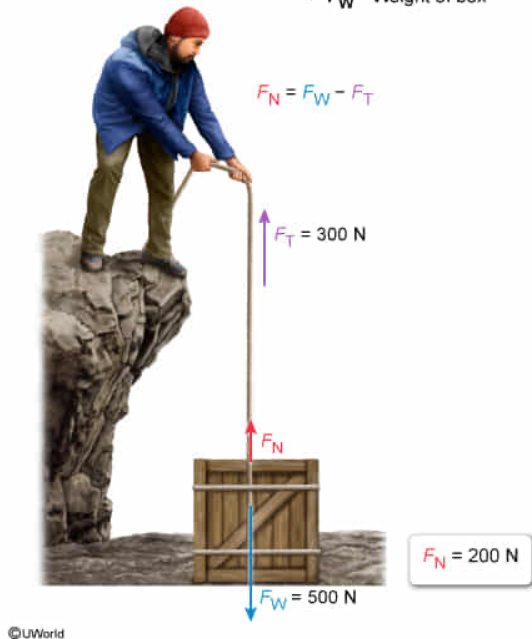
55% answered correctly

Explanation:

Forces on a box in static equilibrium

$$\sum F = 0 = F_T + F_N - F_W$$

- F_T Rope tension
- F_N Normal force on box
- F_W Weight of box



An object is in **mechanical equilibrium** when all forces F acting on it sum to zero:

$$\sum F = 0$$

Mechanical equilibrium includes both **static equilibrium**, in which the object is at rest, and **dynamic equilibrium**, in which the object moves with a constant nonzero velocity. A **free-body diagram** can be used to analyze the forces acting on the object.

In this question, the mass of the box equals 50 kg. Because the **weight** F_W of an object equals the product of its mass m and gravitational acceleration g , F_W of the box equals:

$$F_W = mg = (50 \text{ kg})\left(10 \frac{\text{m}}{\text{s}^2}\right) = 500 \text{ N}$$

The **tension** F_T on the rope is given as 300 N, which is less than F_W . Therefore, the box is resting on the ground in static equilibrium with zero velocity. The forces acting on the box are F_W , F_T , and the **normal force** F_N the ground exerts on the box. These forces all act in the vertical direction, but F_T and F_N are upward whereas F_W is downward. To maintain equilibrium, these forces must sum to zero:

$$\sum F = F_T + F_N - F_W = 0$$

Rearranging this equation to solve for F_N yields:

$$F_N = F_W - F_T = 500 \text{ N} - 300 \text{ N} = 200 \text{ N}$$

Therefore, the magnitude of F_N on the box equals 200 N.

(Choice A) The weight of the box is 500 N, but the tension on the rope reduces the normal force on the box to 200 N.

(Choice B) The tension on the rope is 300 N, but this does not equal the normal force on the box.

(Choice D) The normal force on the box is not 0 N because the tension on the rope is not sufficient to raise the box off the ground.

Educational objective:

Static equilibrium refers to an object at rest where all the forces acting on the object sum to zero. A free-body diagram can be used to analyze forces acting in only one direction or in multiple directions.

Time spent:0 sec

Subject:
Physics

QID:403945

System:
Fluids